

Key Insights

Python + SQL Project

(ITSM Service Desk Operations Analytics Dataset)

Key findings derived from the completed Exploratory Data Analysis.

Analysis Dataset	360,000 SLA-level records
Analysis Columns	29
Period	April 2021 – July 2021
Main Focus	SLA performance, service time and customer satisfaction

1. Overview of Key Insights

The EDA was used to identify the main patterns in the ITSM ticket and SLA data. The strongest findings are related to SLA performance, actual handling time and the difference between First Response and Resolution SLAs.

The findings below are based on the actual project data and the completed Python analysis. They are written as business observations rather than simply repeating statistical outputs.

2. SLA Breaches Are the Main Performance Concern

The clearest pattern in the analysis is the large difference between SLA records that were Met and those that were Breached.

- There were **310,006 Breached SLA records** compared with **49,994 Met records**.
- Breached records had an average actual time of about **3,535.93 minutes**.
- Met records had an average actual time of about **62.50 minutes**.
- The average SLA variance was about **+2,649.50 minutes for Breached records** and **-57.50 minutes for Met records**.

This shows that SLA breaches are not small differences from the target. The breached records are substantially above their defined SLA target.

3. Resolution SLA Performance Requires Attention

Resolution SLA performance is the most important issue identified in the project.

- The dataset contains **180,000 First Response SLA records** and **180,000 Resolution SLA records**.
- All **180,000 Resolution SLA records** are marked as **Breached**.
- The average actual time for Resolution SLA records is **5,893 minutes**.
- The average SLA variance for Resolution SLA records is **4,453 minutes**.
- Average Resolution SLA performance is about **409.24% of the target**.

This pattern suggests that the resolution stage should be investigated first when looking for opportunities to improve overall SLA performance.

4. First Response Performance Is Comparatively Better

First Response SLA records show both successful and breached outcomes, which makes them useful for comparison.

First Response Status	Average Actual Time	Average Performance
Met	62.50 minutes	52.08%
Breached	272.46 minutes	227.05%

The difference between Met and Breached First Response records indicates that response-time targets are achievable in at least part of the data, while breached cases need further investigation.

5. Customer Satisfaction Does Not Show a Strong Difference by SLA Status

Customer satisfaction was also compared with SLA performance. The average CSAT was almost the same for both SLA statuses.

SLA Status	Average CSAT	Average Actual Time	Average SLA Performance
Breached	3.008	3,535.03 min	332.82%
Met	3.012	62.50 min	52.08%

The average CSAT is approximately **3.01 for both groups**. The current analysis therefore does not support a strong claim that SLA status alone determines customer satisfaction.

6. Resolution Code Has Limited Effect on the Major Time Difference

Eight resolution codes were available in the data. Their record counts were reasonably similar, but the average actual times were also very close to one another.

- **Access Granted:** approximately 3,053.60 minutes.
- **Configuration Updated:** approximately 3,053.76 minutes.
- **Known Error Workaround:** approximately 3,053.39 minutes.
- **Password Reset Completed:** approximately 3,053.93 minutes.
- **Patch Applied:** approximately 3,053.77 minutes.
- **Resolved by KB Article:** approximately 3,053.55 minutes.
- **User Educated:** approximately 3,052.86 minutes.
- **Vendor Fix Applied:** approximately 3,053.72 minutes.

The differences are very small. Therefore, resolution code does not appear to explain the major SLA time difference seen between Met and Breached records.

7. Data Quality and Analysis Limitation

An important limitation was found during data exploration: several ticket-level categorical variables contain only one category in the analysed data.

Variable	Available Category
Ticket type	Incident
Category	Software
Sub-category	Patch Update
Priority	P3
Impact	Medium
Urgency	Medium
Ticket status	Closed
Source channel	Email
Created month	June 2021

Because these fields do not contain enough variation, comparisons such as **Priority vs Ticket Status** or **Category vs Ticket Status** cannot provide meaningful business conclusions. These analyses were therefore not treated as key findings.

8. Importance of the Derived SLA Measures

Two additional measures were created during data preparation to make SLA performance easier to interpret.

Derived Column	Meaning	Main Use
sla_variance_minutes	Actual time minus SLA target time	Shows how far performance is above or below the target.
sla_performance_percentage	Actual time as a percentage of SLA target	Makes SLA performance easier to compare across records.

9. Business Implications

- Resolution SLA performance should be reviewed as the first priority because all Resolution SLA records in the analysed data are breached.
- Breached cases should be examined to understand where the additional handling time is being introduced.

- First Response performance can be used as a reference point because it contains both Met and Breached outcomes.
- Customer satisfaction should continue to be monitored, but the current data does not justify claiming a strong direct relationship between SLA status and average CSAT.
- Future datasets should contain more variation in ticket categories, priorities, statuses and channels to support stronger business comparisons.

10. Key Recommendations

Investigate Resolution SLA Breaches

Review the workflow from ticket assignment to final resolution and identify where delays occur.

Monitor SLA Variance

Use the difference between actual and target time to identify cases with large delays.

Track First Response and Resolution Separately

The two SLA types behave very differently, so combining them can hide important performance differences.

Continue Monitoring Customer Feedback

CSAT should remain a KPI even though the current analysis does not show a strong difference by SLA status.

Improve Data Variety in Future Analysis

More diverse ticket categories and statuses would allow stronger comparisons and more detailed business insights.

11. Final Key Takeaway

The main message from the EDA is that SLA performance is the central issue in the analysed ITSM data. The strongest concern is Resolution SLA performance, where all records are marked as breached and average actual time is far above the target.

At the same time, customer satisfaction remains close to **3.01 on average for both Met and Breached records**, so SLA performance and customer satisfaction should not be treated as directly equivalent based on this dataset alone.

Overall, the findings provide a practical starting point for improving resolution processes, monitoring SLA delays and improving the quality and variety of future ITSM data.

THANKYOU!!!!
